Restoring Equivalent Exchange

A Reductive Currency System

Proof of Return

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All things are impermanent — everything arises, exists, and returns.

Abstract

Everything that humans exchange daily — labor, time, meals, services — is finite and returns the moment it is consumed. Yet the medium designed to exchange these things — currency — has been engineered not to return. Gold does not decay. Bitcoin is preserved permanently at 21 million units.

This asymmetry has bred hoarding, obstructed circulation, and entrenched a structure in which only those closest to the faucet benefit (the Cantillon effect). The root cause of 150 years of structural defects in the financial system is a single contradiction: the value being exchanged returns, yet the medium of exchange does not.

This paper proposes a currency system that resolves this contradiction by embedding return into currency itself. The system comprises three currencies: Base, which guarantees existence; Amortization, which is Base returned through exponential decay upon use; and Creation, which supports creative endeavor — the initials of Amortization, Base, and Creation form A, B, C. The core is an automated currency issuance and return system driven by an exponential model. Currency is issued, used, and returned according to natural law. Through the same exponential feedback found in nature — radioactive decay, thermal cooling, ecosystem cycles — issuance and return maintain autonomous equilibrium without human intervention.

Satoshi Nakamoto demonstrated through Proof of Work “a method for reaching consensus without a trusted third party.” This paper demonstrates through Proof of Return “a method for exchanging things that return, with a medium that also returns.” The entire system is described by just two exponential functions — decay and recovery.

1. Broken Money — Why Currency Stopped Working

In Broken Money (2023), Lyn Alden provided a comprehensive diagnosis of the structural defects in the modern financial system. Since the invention of the telegraph in the 1850s, transactions could move at the speed of light, but settlement of physical assets like gold could only move at the speed of matter. Banks monopolized this gap, and the central banking system was born. Currency unpegged from gold can expand infinitely at the discretion of its managers. Expanded currency does not spread evenly — a structure is entrenched in which those closest to the faucet receive it first.

1.1 The Cantillon Effect — Those Closest to the Faucet Benefit

Newly created currency does not spread evenly. Those closest to the source (central banks, governments) receive it first; by the time it reaches the periphery, purchasing power has already been diluted. This phenomenon, described by Richard Cantillon in the 18th century, has only accelerated in the modern era. Between 2020 and 2022, the net worth of the top 1% in the United States increased by $11.8 trillion. The bottom 50% gained only $1.5 trillion.

This structure is not intentional design but a side effect of currency that “does not disappear.” Currency that does not disappear can be hoarded. Currency that can be hoarded concentrates

among those closest to the faucet. Concentrated currency becomes the basis for further credit creation, self-amplifying inequality. Central banks issue currency, commercial banks multiply credit, expanded credit inflates asset prices, and inflated assets further swell the net worth of asset holders. The wages received by workers at the periphery sit at the end of this chain. By the time currency reaches them, purchasing power has already been stripped.

This does not stop at national borders. Developed nations hold debt in their own currencies, but developing nations bear debt in foreign currencies (primarily the US dollar). Every time the dollar tightens, citizens of Lebanon, Turkey, and Argentina find themselves unable to withdraw their own deposits. The Cantillon effect crosses borders, stripping purchasing power from the periphery on a global scale.

1.2 Financialization — The Collateral Damage of Currency's Defect

In an environment where currency continually depreciates, people flee to every scarce asset, loading a "monetary premium" onto each one. Housing, equities, and collectibles become speculative instruments, diverging from their intrinsic use-value. Soaring housing prices exclude the middle class, and with each economic crisis, debt is never fully reset — the starting point of the next cycle is always higher. The structural defects of the financial system are rooted in both economic inequality and the retreat of political freedom.

2. The Root Cause — The Collapse of Equivalent Exchange

The structural defects described in the preceding chapter derive from a single root cause.

Everything that humans exchange daily returns.

Labor ends. Meals are consumed. Time passes. Services are complete the moment they are received. Caregiving, education, creation — all of these exist, generate value, and return. The producers themselves are also finite. A person is born, creates value, and dies. Credit creation springs from trust in a person. When that person dies, the credit born from them should, by nature, also return.

Yet the medium of exchange — currency — has been designed not to return.

In the current financial system, when a person dies, their debt persists. It is inherited, securitized, and resold. The person has returned, yet the credit alone drifts on forever. This is the root cause of debt accumulation.

Gold does not decay. Banknotes are intentionally made durable. Bitcoin is designed to be preserved permanently at 21 million units. Satoshi Nakamoto solved "how to reach consensus without a trusted third party." But the property that "currency can be stored" was left intact — in fact, reinforced. As a result, Bitcoin was consumed by capitalism. HODL (hoarding) became its culture, and it became a speculative asset.

2.1 The Principle of Equivalent Exchange

Equivalent exchange means exchanging things of the same nature. If what is being exchanged returns (labor, meals, time), then the medium of exchange must also return — otherwise it is not equivalent.

The current currency system violates this principle. It exchanges things that return with a medium that does not. As a result, value accumulates on the side of the medium (currency), and is stripped from the original objects of exchange (labor, creation, life). This is the essence of financialization and the origin of the Cantillon effect.

2.2 Impermanence and Currency Design

"Nothing remains unchanged; everything is in constant flux."

Subatomic particles appear and vanish. Stars burn and return. People are born and die. Ecosystems cycle. At every scale of the universe, existence follows a cycle of arising, existing, and returning.

In Broken Money Chapter 4, Alden observed that human-managed ledgers see entropy increase ceaselessly, like the second law of thermodynamics. She described this as a problem. But the increase of entropy is not a problem — it is the law of the universe itself. The problem is having designed something that should return not to return.

Embedding return into currency is not a repair of defects. It is alignment with natural law.

3. The Proposed System — ABC Reductive Currency

This system comprises three currencies — Amortization, Base, and Creation. It is a "principle-embedded currency" in which the exponential model is built into the currency itself. In addition, a Stablecoin Pool (SCP) serves as the interface with external currencies, but it is not a currency — it is the backing fund.

3.1 Base (Existence Guarantee Currency)

The foundational currency guaranteeing survival. The ceiling for Base is each participant's deposit amount, up to a maximum of 1,000 Base (= 1,000 USDT). Base is held in the individual's Base wallet. The used portion recovers exponentially — faster when the balance is low, more gradually as it approaches the ceiling — a guarantee of existence itself.

The instant a payment is made from the Base wallet, Base becomes Amortization and enters the recipient's Amortization wallet. What was an existence guarantee completes its role as a medium of exchange and enters the cycle of return — this is the transformation from Base to Amortization.

3.2 Amortization (Return Currency)

All three currencies received from others enter the Amortization wallet, where they are continuously returned through an exponential decay model, reaching a 3% residual rate after

108 days.

In nature, ash after combustion retains 1–3% of its original mass and returns to the soil. Cremated remains weigh 3–5% of the body. Fallen leaves retain 5–10% as humus after decomposition, becoming the soil for the next forest. The 3% residual rate is grounded in nature's residue rates.

If residue is used in a transaction, it merges with other currency in the recipient's Amortization wallet and lives out a new life. If unused, it asymptotically approaches zero. Currency that has decayed to a negligible residual naturally returns to dust. This is the natural form of return. In nature, ash is not cleaned up by a designer — time returns it.

Payments can also be made from the Amortization wallet. The paid amount enters the recipient's Amortization wallet, where return continues at the same decay rate.

Currency in the Amortization wallet cannot be exchanged for stablecoins. By design, no escape route exists to convert a returning currency into an asset that "does not disappear."

3.3 Creation (Creative Support Currency)

A currency for education, entrepreneurship, creation, and research. It is activated when the system reaches a certain scale (10 million participants or more). Based on participant applications, up to 100,000 Base (USDT equivalent) is granted. The grant frequency and amount are determined through deliberation between participants and Decentralized AI. Applications are evaluated through deliberation between participants and Decentralized AI. The implementation of this process is determined by participants at the maturation stage of the economic zone. Spent Creation also enters the Amortization wallet and is returned through the same exponential decay model.

Simulations have confirmed that the same single equation as Amortization produces the same autonomous equilibrium for Creation currency. Exponential decay converges the entire system to a steady state at the same speed.

3.4 Stablecoins as the Entry Point

Upon joining, participants convert USDT (or USDC) into Base. The conversion ceiling is 1,000 Base (= 1,000 USDT). Stablecoins deposited at entry are fully stocked in the Stablecoin Pool (SCP). Amounts exceeding the 1,000 USDT Base ceiling are also accumulated in the SCP.

1 Base = 1 USDT is merely the entry peg. As the economic zone expands, the significance of this peg in internal transactions becomes vestigial.

3.5 Stablecoin Pool (SCP)

In this system, the SCP is not a currency. It is the connection point with existing financial regulations and the backing fund of this economic zone.

Participants can fully withdraw their SCP deposits at any time — no lock-up period is imposed.

The management policy of the SCP is decided by DAO vote of participants and Decentralized AI when the economic zone has fully matured. Until then, it is safely stored as USDT. The designer does not predetermine the use of SCP funds.

The SCP terminates its role when the economic zone achieves complete independence from the existing capitalist system.

4. Currency Architecture — The Exponential Model

Satoshi Nakamoto's Bitcoin paper (2008) answered the question "How can strangers reach consensus without a trusted third party?" through a mechanism called Proof of Work.

This system introduces the concept of Proof of Return. It makes verifiable on the blockchain the fact that currency has returned — the process by which all issued currency returns, through exponential decay, without human intervention. Where Proof of Work proves "computational resources were expended," Proof of Return proves "currency was returned in accordance with natural law."

4.1 Two Exponential Functions

The Amortization wallet is the destination for all three currencies received from others. The balance is returned at the same decay rate every second. The decay function: Balance(t+1 second) = Balance(t) × (1 − r)

The Base wallet is the vessel for a participant's "existence guarantee." Used Base recovers exponentially. The recovery function: Recovery = (deposit amount − current balance) × 6% (daily)

In the Amortization wallet, exponential decay drives the balance toward 0. In the Base wallet, exponential recovery drives the balance toward the deposit amount. The same mathematics works in opposite directions. Decay and recovery are mirror images.

4.2 Amortization — Return Through Exponential Decay

The core of the Amortization wallet is continuous return through exponential decay.

Balance(t+1 second) = Balance(t) × (1 − r)

This single equation governs the entire currency system. The per-second decay rate r is set so that the residual rate reaches 3% at the internal constant T = 108 days.

Daily decay rate: r(day) = 1 − 0.03^(1/108) ≈ 3.174%

108 days equals 3.55068 months. On the surface of the paper, we write "T = 3.6 months"; in the smart contract, we inscribe T = 108 days. The 3% residual rate is based on the natural law described above.

There are three reasons for adopting exponential decay.

First, return in nature is almost entirely exponential. Radioactive decay, thermal cooling, ecosystem balance — the more there is, the more decays; the less there is, the less decays. It does not mean “one step down every day at midnight” — everything is in constant flux.

Second, exponential decay is self-regulating. Even as participants grow, usage rates change, and turnover rises, issuance and return autonomously equilibrate. Simulations have confirmed that the system converges to a steady state across all parameter combinations.

Third, batch management becomes unnecessary. With exponential decay, the same rate applies to the entire wallet balance, so regardless of when a deposit was made, it can be processed with a single formula: balance × rate.

4.3 Base — Existence Guarantee Through Exponential Recovery

Recovery of the Base wallet also follows an exponential function. The used portion recovers exponentially — faster when the balance is low, more gradually as it approaches the ceiling.

Recovery = (deposit amount − current balance) × 6% (daily)

If the deposit amount is 1,000 and the balance is 100, then (1,000 − 100) × 0.06 = 54 Base recovers. If the balance is 900, then (1,000 − 900) × 0.06 = 6 Base recovers. The closer to the ceiling, the slower the recovery, and it never exceeds the deposit amount. Even from a zero balance, as long as USDT is deposited in the SCP, approximately 37 days restores 90%. This is calculated per-second, just like Amortization. Even when the balance reaches zero, something invisible remains — just as an exponential function theoretically never reaches zero. A person can always start over.

4.4 Implementation of Continuous Return — Lazy Evaluation

Smart contracts employ lazy evaluation. Only the last update timestamp and the balance are recorded; when a transaction occurs, the current residual value is computed from the elapsed seconds. It does not “run a calculation every second.”

4.5 Design Parameters

Parameter	Value	Notes
Return model	Exponential decay	Balance(t+1s) = Balance(t) × (1−r)
Internal constant T	108 days	The number of earthly desires. Written as 3.6 months
Residual rate at 108 days	3%	The number of generation. Based on nature’s residue rates
Daily decay rate	≈ 3.174%	r = 1 − 0.03^(1/108)
Base deposit ceiling	1,000 USDT	Excess goes to SCP
Base recovery model	Exponential recovery 6% (daily)	Recovery = (deposit amount − balance) × 6%/day

At zero balance	Recovers (≈37 days to 90%)	Recovers as long as SCP deposit exists
1 Base	= 1 USDT	Entry peg only
Creation activation	10 million participants+	After economic zone independence

4.6 Constants: 3-6-9

The constants of this system are 3 and 6. The Amortization residual rate is 3%. The recovery rate is 6%. The entire system is described by just two exponential functions: the decay of Amortization (3%) and the recovery of Base (6%). The breathing of the universe — exhaling and inhaling — directly implemented.

Three is the smallest number at which generation occurs. Six is the smallest perfect number (a number equal to the sum of its proper divisors). 108 is the number of earthly desires in Buddhism.

108 ÷ 3 = 36. 36 ÷ 6 = 6. The digital root of 108 is 9 (1+0+8); the digital root of 36 is also 9 (3+6). Converting 108 days to months yields 3.55068 months — its digital root is again 9 (3+5+5+0+6+8 = 27 → 9). T = 3.6 is also 9.

Three and six appear on the surface; nine pervades the entire structure.

5. Implementation

5.1 Technical Foundation

This system is implemented as smart contracts on an Ethereum Layer 2 (L2) network. The choice of L2 is based on the cryptocurrency trade-offs organized by Alden in Chapter 22. Improving base layer throughput increases node operation costs and compromises network decentralization. L2 decouples throughput from L1 while relying on L1 for security, resolving this trade-off.

Initial deployment uses Base network (an L2 operated by Coinbase). Gas fees are extremely low, enabling the continuous calculations of exponential decay and Amortization wallet balance updates to operate at practical cost. It is a suitable environment for launching operations with initial funding.

However, Base network is a chain operated by a single company, Coinbase, and is not the final form. As the economic zone grows, the system will migrate to a more neutral L2, or to a dedicated L2 (rollup) that inherits Ethereum's security. Security is anchored to the Ethereum mainnet; daily transactions run cheaply and quickly on L2. It is the same structure as an independent nation sharing a security alliance while maintaining sovereignty.

5.2 Three Layers of Transparency

Contract layer: Fund management is handled by smart contracts. Neither individuals nor

operators can directly manipulate funds. The code is published as open source and undergoes external audits.

Visibility layer: Deposit and withdrawal amounts, return history, and SCP balances are fully disclosed on-chain. A state of real-time verifiability by anyone is maintained.

Freedom of withdrawal: Participants can withdraw their assets at any time. No lock-up period is imposed.

5.3 Staged Deployment

Phase 1 (Launch): Initial funding by the author. Deployment of smart contracts on Base network. Recruitment of external participants through paper publication.

Phase 2 (Economic activity): First transactions between participants. The return cycle of exponential decay begins, and the system draws its first breath.

Phase 3 (Scale expansion): Increased participant inflow. Beginning of active economic activity.

Phase 4 (Full deployment): 10 million participant scale. Activation of Creation currency issuance. Migration to a more neutral L2 or a dedicated L2 inheriting Ethereum's security. Independence of the economic zone.

6. Relation to Prior Work

6.1 The Matsuda Plan — The Origin of Inspiration

The Matsuda Plan is a proposal for resolving the national debt problem and realizing a basic income through government-issued digital currency. It was the origin of inspiration for this system, and the idea of separating "a currency for survival guarantee" from "a currency for challenge" was sparked by the Matsuda Plan.

However, this system has arrived at a fundamentally different design from the Matsuda Plan. First, the Matsuda Plan envisions a currency issued and managed by the state, whereas this system has no issuer. Smart contracts operate autonomously, eliminating arbitrary human judgment. Second, the Matsuda Plan's currency, like existing fiat currency, is designed "not to disappear," whereas this system's currency is returned according to natural law through exponential decay. Third, the Matsuda Plan's three-layer structure is an administrative classification, whereas ABC Reductive Currency's three layers are a functional separation based on the physical properties of currency — existence guarantee (Base), return (Amortization), and creative support (Creation).

The question the Matsuda Plan raised — "Separate currency by function" — was correct. This paper offers an answer to that question based not on the state, but on natural law.

6.2 Broken Money — Diagnosis Separated from Prescription

Alden's Broken Money is the most comprehensive diagnostic report on why the modern financial system is broken. The gap between transaction speed and settlement speed, the Cantillon effect, long-term debt cycles, financialization, the export of disruption to the periphery — these analyses form the foundation for this paper's problem statement.

However, Alden did not write a prescription. ABC Reductive Currency, the homeostatic model of exponential decay, and the concept of Proof of Return are original proposals of this paper. Building on Alden's diagnosis, this paper presents a structural solution to the root cause — currency that does not return.

6.3 Bitcoin — Solved the Trust Problem, but Not the Storage Problem

Satoshi Nakamoto's Bitcoin (2008) solved "how to reach consensus without a trusted third party." In replacing the ledger's manager from humans to mathematics, it was a landmark invention in the history of currency.

However, Bitcoin did not recognize the "storability" of currency as a problem. Rather, it made its 21 million supply cap and permanent preservation its defining features. As a result, Bitcoin was consumed as a speculative asset by the existing capitalist structure. As HODL culture demonstrates, storability creates an incentive to hoard, causing the original exchange function of currency to recede.

To the question Bitcoin left behind — "Why use a medium that does not return to transact things that do?" — this paper offers the return mechanism of exponential decay as its answer.

7. Conclusion

Everything arises, exists, and returns.

Subatomic particles appear and return. Stars burn and return. People are born and die. Labor ends, meals are consumed, time passes. At every scale of the universe, existence has a cycle of return.

Currency alone has been severed from this cycle.

Gold does not return. Bitcoin does not return. Fiat currency increases or decreases only by human judgment. As a result, an asymmetry emerged: exchanging things that return with a medium that does not. This asymmetry has braked economic activity.

This paper resolves this asymmetry by embedding return into currency. The only equation inscribed in the smart contract is: balance × (1−r). The internal constant T = 108 days is inscribed as the number of earthly desires, and after 108 days, 3% of residue remains as the seed of the next cycle. All other parameters fluctuate naturally according to usage rates. No human controller is needed; the system breathes autonomously.

Radioactive decay, thermal cooling, ecosystem balance — return in the natural world is entirely governed by exponential feedback. The more there is, the more decays; the less there is, the

less decays. That is why it equilibrates. This system did not design a currency. It translated natural law into currency.

Exchange things that return, with a medium that returns.

This is equivalent exchange.

This system is hereby named:

AMI

All Material is Impermanent

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— Choose from love, not from fear. —